

Difference-in-Differences with Multiple Time Periods*

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Abstract

In this article, we consider identification, estimation, and inference procedures for treatment effect parameters using Difference-in-Differences (DiD) with (i) multiple time periods, (ii) variation in treatment timing, and (iii) when the “parallel trends assumption” holds potentially only after conditioning on observed covariates. We show that a family of causal effect parameters are identified in staggered DiD setups, even if differences in observed characteristics create non-parallel outcome dynamics between groups. Our identification results allow one to use outcome regression, inverse probability weighting, or doubly-robust estimands. We also propose different aggregation schemes that can be used to highlight treatment effect heterogeneity across different dimensions as well as to summarize the overall effect of participating in the treatment. We establish the asymptotic properties of the proposed estimators and prove the validity of a computationally convenient bootstrap procedure to conduct asymptotically valid simultaneous (instead of pointwise) inference. Finally, we illustrate the relevance of our proposed tools by analyzing the effect of the minimum wage on teen employment from 2001–2007. Open-source software is available for implementing the proposed methods.

JEL: C14, C21, C23, J23, J38.

Keywords: Difference-in-Differences, Dynamic Treatment Effects, Doubly Robust, Event Study, Variation in Treatment Timing, Treatment Effect Heterogeneity, Semi-Parametric.

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that are potentially associated with the outcome evolution of $Y(0)$ during post-treatment periods. We explicitly rule out incorporating post-treatment covariates as they can potentially be affected by the treatment; see, e.g., [Wooldridge \(2005b\)](#), for a related discussion under the unconfoundedness setup.

3 Summarizing Group-Time Average Treatment Effects

The previous section shows that we can identify the $ATT(g, t)$'s by restricting treatment anticipation behavior and imposing a conditional parallel trends assumption. In many applications, the $ATT(g, t)$'s can be the ultimate causal parameters of interest. They can be used to highlight treatment effect heterogeneity across different groups g , at different points in time t , and across different lengths of treatment exposure, $e = t - g$. In other situations, however, researchers may want to combine these different $ATT(g, t)$'s to form more aggregated causal parameters. For instance, if the number of groups and time periods is relatively large, it may be challenging to interpret many group-time average treatment effects.

In this section, we consider different aggregation schemes for the $ATT(g, t)$'s that allow researchers to form a variety of summary measures of the causal effects of a given policy. Our aggregation schemes are of the form

$$\theta = \sum_{g \in \mathcal{G}} \sum_{t=2}^{\tau} w(g, t) \cdot ATT(g, t), \quad (3.1)$$

where $w(g, t)$ are carefully-chosen (known or estimable) weighting functions specified by the researcher such that θ can be used to address a well-posed empirical/policy question. Different choices of $w(g, t)$ allows researchers to highlight different types of treatment effect heterogeneity. We pay particular attention to aggregations that result in a single overall treatment effect summary parameter as well as to aggregations related to understanding dynamic effects as is commonly done in event-study analysis. Of course, many other aggregated parameters of the type (3.1) can be easily constructed following our framework. We illustrate this point by also summarizing heterogeneity with respect to group or by calendar time.

Before proceeding with the discussion on how to construct these different aggregated parameters, it is worth revisiting the two most popular treatment effect summary measures used by practitioners in DiD setups. These are based on the “static” and “dynamic” two-way fixed effects (TWFE) linear regression specifications

$$Y_{i,t} = \alpha_t + \alpha_g + \beta D_{i,t} + \epsilon_{i,t}, \quad (3.2)$$

$$Y_{i,t} = \alpha_t + \alpha_g + \sum_{e=-K}^{-2} \delta_e^{anticip} \cdot D_{i,t}^e + \sum_{e=0}^L \beta_e \cdot D_{i,t}^e + v_{i,t}, \quad (3.3)$$

respectively, where α_t is a time fixed effect, α_g is a group fixed effect, $\epsilon_{i,t}$ and $v_{i,t}$ are error terms, $D_{i,t}^e = 1\{t - G_i = e\}$ is an indicator for unit i being e periods away from initial treatment at time t , and K and L are positive constants. The parameter of interest in the static TWFE specification is β , which, in applications, is typically interpreted as an overall effect of participating in the treatment across groups and time periods. In the dynamic TWFE specification, practitioners usually focus on the β_e , $e \geq 0$, and these parameters are typically interpreted as measuring the effect of participating in the

treatment at different lengths of exposure to the treatment.

Despite the popularity of these specifications, recent research has shown that one must be very careful in attaching a causal interpretation to these aggregated parameters. For instance, [Borusyak and Jaravel \(2017\)](#), [Goodman-Bacon \(2019\)](#), [de Chaisemartin and D’Haultfœuille \(2020\)](#), and [Athey and Imbens \(2018\)](#) have shown that, in general, β recovers a weighted average of some underlying treatment effect parameters but some of the weights on these parameters can be negative. This can potentially lead to particularly problematic cases such as the effect of the treatment being positive for all units, but the TWFE estimation resulting in estimates of β that are negative. Even in cases where the weights are not negative, the weights on underlying treatment effect parameters are entirely driven by the TWFE estimation strategy and are sensitive to the size of each group, the timing of treatment, and the total number of time periods (see Theorem 1 in [Goodman-Bacon \(2019\)](#)). The results in this section can be used in exactly the same setup to identify a single interpretable average treatment effect parameter and, thus, provide a way to circumvent the issues with the more common approach.

As discussed by [Goodman-Bacon \(2019\)](#), the “negative weight problem” associated with β arises when treatment effects evolve over time. Thus, one may wonder if such problems would still be present when considering more general, dynamic specifications such as (3.3). [Sun and Abraham \(2020\)](#) shows that this is still the case as the β'_e s associated with (3.3) do not recover easy-to-interpret causal parameters and still generally suffer from the same sorts of “negative weighting problems.” In contrast to this, we provide a simple way to directly aggregate our group-time average treatment effects into average treatment effects across different lengths of exposure to the treatment.

3.1 Aggregations to Highlight Treatment Effect Heterogeneity

Next, we discuss several partial aggregations of the group-time average treatment effects in order to summarize different dimensions of treatment effect heterogeneity. Although there are additional possibilities, we focus our discussion below on how to answer three particular questions: (a) How does the effect of participating in the treatment vary with length of exposure to the treatment? (b) Do groups that are treated earlier have, on average, higher/lower average treatment effects than groups that are treated later? (c) What is the cumulative average treatment effect of the policy across all groups until some particular point in time? Throughout this section, to avoid notation clutter, we assume that units do not anticipate treatment, i.e., we consider the case where Assumption 3 holds with $\delta = 0$. We also assume that a “never treated” group is available.

3.1.1 How do average treatment effects vary with length of exposure to the treatment?

One of the most popular questions that arises in DiD setups with multiple time periods concerns treatment effect dynamics: How does the effect of participating in the treatment vary with length of exposure to the treatment? For instance, do average treatment effects increase/decrease with elapsed treatment time? Indeed, answering this type of question is often the main motivation for using the event study regression in (3.3), though, as we mentioned above, that sort of regression may not be suitable for such a task. In this section, we propose an aggregation scheme that is suitable to highlight treatment effect heterogeneity with respect to length of exposure to the treatment that does not suffer from the drawbacks associated